

Dmitriy Yanin

DATA ENGINEER

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Senior engineering leader with 10+ years of experience at the intersection of energy systems, scientific research and, large-scale on-prem/cloud computing. Proven track record building scalable data platforms and leading multidisciplinary engineering teams delivering production systems for battery intelligence and decarbonisation technologies. Combining a PhD in inorganic chemistry with deep expertise in cloud architecture, data engineering, and automation, I translate complex scientific and industrial data challenges into robust, scalable software systems. My work spans large-scale computational materials discovery, AI-assisted battery research, and cloud data platforms powering analytics for automotive and grid-scale energy storage. Experienced in defining platform strategy, leading cross-functional teams, and delivering secure, compliant infrastructure (ISO 27001 / SOC 2) that enables advanced analytics, product development, and large-scale data operations. Passionate about building high-impact engineering teams and data platforms that accelerate innovation in energy and climate technologies.

Work Experience

Checktrade | Data & Insights

Portsmouth, UK

DATA ENGINEER

Jan. 2024 — Current

- Collaborate within a core data engineering team to architect, build, and manage scalable Airflow-based ELT pipelines (Python & SQL) integrated with a Starburst data lakehouse.
- **End-to-End Pipeline Engineering:** Design and execute robust data ingestion, transformation, and export workflows, structuring raw data into a high-performance Medallion architecture (Bronze, Silver, Gold).
- **Multi-Cloud & Ecosystem Integration:** Streamline data collection by centralising diverse data streams from AWS (S3), Microsoft SharePoint, Salesforce, and the Google Cloud ecosystem (including BigQuery, Cloud Storage, Firestore, Cloud SQL, PubSub, and Workspace).
- **AI & LLM Data Preparation:** Process and refine conversation intelligence data from CallMiner to construct high-quality training datasets for internal LLM models.
- **External API & Analytics Data Integration:** Build automated pipelines to ingest and process web and business data from Optimizely (analytics), Google Places API, and an automated data upload process to Google Actions Center API with data validation and issue alerting.
- **Legacy Modernisation:** Spearheaded the architectural realignment of legacy systems into the Medallion framework, improving data quality, governance, and query performance.
- **Code Refactoring & System Upgrades:** Extensive SQL and Python code refactoring to seamlessly support major upgrades to core business systems, including telephony logging, product taxonomy, customer job requests, and printed Trade Directories.
- **Modern Data Stack Innovation:** Evaluated pipeline efficiencies by driving a dbt (Data Build Tool) proof-of-concept to migrate Gold Layer transformations away from Apache Airflow.
- **Web Operations Support:** Developed automated data pipelines to generate critical sitemap files for the core Checktrade website to support SEO health.

Sliide | Data Engineering

London, UK

SENIOR DATA ENGINEER

Aug. 2021 — Jan. 2024

- Collaborate within a core data engineering team to architect and build automated, resilient ETL solutions using Python, SQL, and Jenkins across multi-cloud environments (AWS and GCP).
- **Cloud Migration Leadership:** Served as Technical Lead for a major data warehouse migration from Snowflake to an AWS ecosystem. Successfully designed and executed pipelines utilizing Redshift, S3, DataSync, Apache Airflow, and AWS Glue.
- **Team Mentorship:** Actively mentored and upskilled junior team members during complex platform migrations, fostering technical growth and accelerating project delivery.
- **Data Governance & Lineage:** Designed and implemented a comprehensive, cross-platform data cataloguing solution. Established multi-level data lineage across diverse ecosystems, including Snowflake, BigQuery, Redshift, Tableau, GCS, S3, and Google Sheets.
- **Pipeline Resilience:** Engineered automated, self-healing data pipelines within Airflow and Jenkins, significantly reducing manual intervention and minimizing production downtime.
- **Multi-Source Data Ingestion:** Built scalable ingestion pipelines to process and integrate high-volume data from third-party APIs, PostgreSQL databases, Google Cloud Storage, and AWS S3.
- **Strategic Architecture Evaluation:** Headed a proof-of-concept (PoC) platform migration from Snowflake and Jenkins to Google BigQuery to evaluate cost-efficiency and performance scalability.

Waitrose Ltd. | Ecommerce & Digital Marketing

Bracknell, UK

DATA ENGINEER & ANALYTICS IMPLEMENTATION SPECIALIST

Jan. 2018 — Aug. 2021

- Leader with full ownership of end-to-end web/app analytics and cloud data infrastructure across Waitrose.com, specialist shops, and mobile applications. Partnered with enterprise stakeholders to transform complex business requirements into scalable, secure, and automated multi-cloud data solutions.
- **Enterprise Marketing Automation:** Designed and engineered a GCP-hosted 'Digital Marketing Hub'. This fully automated platform leverages first-party customer data to orchestrate audience segments across Google Ads, Facebook, Salesforce, Monetate, and Google Analytics.
- **Advanced GCP Architecture:** Constructed the Digital Marketing Hub using a modern, serverless cloud stack including App Engine, Cloud Functions, BigQuery, PubSub, KMS, AI Tables, Stackdriver Logging/Monitoring, and Looker Studio.
- **Technical Leadership:** Held complete accountability for the business analysis, system architecture, and core development of the marketing hub, while hiring, training, and leading an engineering team to deliver project modules.
- **Real-Time Data Streaming:** Architected an end-to-end, cross-cloud streaming solution to ingest, process, and analyze real-time AWS Kinesis feeds using Google App Engine, PubSub, Dataflow, BigQuery, and Looker Studio.
- **Data-Driven Ad Automation:** Built a suite of automated algorithmic engines for targeted Google advertising and dynamic search bidding, integrating real-time competitor pricing matrices with historical customer purchase behaviors.
- **AI-Powered Customer Insights:** Pioneered a sentiment and text analysis pipeline using Google Natural Language Processing (NLP) to parse, classify, and extract actionable insights from customer complaints and feedback channels.
- **Advanced Tracking & Tagging:** Scaled the telemetry and analytics capabilities of all e-commerce platforms and mobile apps by deploying advanced, custom Google Tag Manager (GTM) frameworks.
- **Data Governance & GDPR Compliance:** Spearheaded technical GDPR compliance frameworks; built a custom automated cookie-auditing tool and engineered a robust consent management architecture within GTM.
- **Automated ETL & Reporting:** Developed a comprehensive suite of serverless ETL pipelines to automate complex data extractions from BigQuery, streamlining reporting workflows for cross-functional business stakeholders.

Waitrose Ltd. | Technical & CSR

Bracknell, UK

PRODUCT INFORMATION TECHNOLOGIST

May 2008 — Dec. 2017

- Orchestrated project delivery, systems analysis, and third-party vendor management across critical retail and complaint management platforms. Functioned as an enterprise super-user, engineering advanced reporting solutions, optimizing data workflows, and enforcing rigorous data quality standards across legacy and modern systems.
- **Business-Critical Regulatory Delivery:** Led the technical execution of a high-priority compliance project for European Food Information Regulations. Partnered with the Legal team to define system requirements and engineered a cross-platform solution linking third-party technical systems, in-house data storage, ETL workflows, and the main e-commerce platform.
- **Business Intelligence (BI) Modernisation:** Spearheaded a reporting transformation by pioneering the adoption of Business Objects. Designed a comprehensive suite of enterprise reports and established a sustainable data culture by training and mentoring a cohort of system super-users.
- **Workflow Automation & Scripting:** Engineered numerous automated dashboards and reporting pipelines by designing custom analysis, scheduling, and ETL tools utilizing VBA, Google Apps Script, and C#.
- **Cloud Innovation:** Designed and deployed a fully automated, cloud-based product complaint ingestion feed leveraging Google Cloud Platform (GCP).
- **Product Development & Risk Assessment:** Directed the end-to-end delivery of a supplier technical risk assessment platform. Gathered cross-functional requirements, designed top-level architectures and process workflows, and managed senior stakeholder alignment.
- **Cloud Infrastructure Evaluation:** Initiated and managed a proof-of-concept trial utilizing Microsoft Azure environments to evaluate alternative data storage, processing, and analytics capabilities.

Waitrose Ltd. | Distribution

Bracknell, UK

PROJECT IMPLEMENTATION AGENT

Oct. 2003 — May 2008

- **Software Solution Delivery:** Played a major role in the end-to-end delivery and lifecycle management of business-critical software solutions tailored for warehouse operations.
- **User-Centric Product Development:** Drove continuous product improvements by establishing customer feedback loops, capturing technical requirements, and translating them into actionable feature roadmaps.
- **Quality Assurance & Testing:** Directed software troubleshooting efforts and executed rigorous incremental version testing to ensure system stability and minimal deployment risk.
- **Change Management & Enablement:** Authored a comprehensive end-user training curriculum and facilitated classroom-based training sessions, accelerating software adoption across diverse user groups.
- **Proprietary Software Development:** Designed, engineered, and deployed a custom warehouse workforce management software tool from scratch.

Education

University of Reading

Reading, UK

M.Sc. IN NETWORK-CENTERED COMPUTING

Oct. 2008 - May 2010

South-Kazakhstan State University

Shymkent, Kazakhstan

B.Sc. IN AUTOMOTIVE TRANSPORT

Oct. 1993 - Jul. 1998

Skills

Cloud Architecture & Strategy

AI, Machine Learning & LLMs

Data Engineering & Orchestration

Data Warehousing & Analytics Infrastructure

Software Engineering & Workflow Automation

Business Intelligence & Reporting

Leadership & Delivery Methodologies

- Cloud Architecture: Google Cloud Platform (GCP) enterprise solution design.
- Data Migration: Legacy system migrations, data cleansing, schema evolution.
- System Design: Structural software design, cross-platform architecture patterns.
- Generative AI Frameworks: LangChain, Ollama, local model deployment, prompt engineering.
- Cloud AI Platforms: Vertex AI, Google BigQuery AI, Google Natural Language Processing (NLP).
- Predictive Modeling: Core Python machine learning libraries, automated feature preparation.
- ETL/ELT Pipelines: Apache Airflow, dbt, Jenkins, AWS Glue, Google Cloud Composer, Dataflow, Cloud Functions.
- Data Formats: Optimization using Parquet, JSON, XML, CSV exchange layouts.
- API Integration: High-performance REST, GraphQL, and gRPC endpoints in Python and Node.js.
- Data Platforms: Google BigQuery, Snowflake, Amazon Redshift, Trino, Starburst Data Lakehouse.
- Database Design: Advanced data modeling, query optimization, analytics automation.
- Telemetry & Tracking: Google Tag Manager (GTM), Google Analytics 360 (User-level & APIs), Firebase, Optimizely.
- Core Languages: Python, SQL, Node.js, JavaScript, Java.
- Enterprise Automation: Advanced Google Workspace automation, Google Apps Script, Expert MS Office VBA development.
- Web Systems: Scripting, reporting automation, basic front-end tools (HTML).
- BI Platforms: Google Looker Studio, Business Objects dashboards.
- Data Delivery: Automated scheduled distributions to cross-functional stakeholders.
- Business Analysis: Cross-functional requirements definition, systems analysis.
- Project Delivery: End-to-end project management, third-party vendor coordination.
- People Leadership: Strategic stakeholder management, team supervision, technical mentoring, and upskilling.